

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250268746

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Kowalchuk; Jason Ronald et al.

TEMPERATURE THERAPY GARMENT AND METHOD

Abstract

A cold therapy garment formed from one or more clothing panels. Each clothing panel extends from a first edge to a second edge opposite the first edge and includes an integrated flexible gel sheet to be cooled prior to a user wearing the cold therapy garment, each integrated flexible gel sheet extending continuously across the clothing panel from the first edge to the second edge, and each integrated flexible gel sheet formed of a cold-therapy material having a semi-rigid form throughout a selected temperature use range.

Inventors: Kowalchuk; Jason Ronald (London, CA), Beer; Luisa (London, CA)

Applicant: Kowalchuk; Jason Ronald (London, CA); Beer; Luisa (London, CA)

Family ID: 1000007712567

Appl. No.: 18/586200

Filed: February 23, 2024

Publication Classification

Int. Cl.: A61F7/02 (20060101); A61F5/30 (20060101)

U.S. Cl.:

CPC A61F7/02 (20130101); A61F5/30 (20130101); A61F2007/0219 (20130101);
A61F2007/0234 (20130101); A61F2007/0236 (20130101)

Background/Summary

TECHNICAL FIELD

[0001] The disclosure relates to temperature therapy, and particularly to a temperature therapy

garment.

BACKGROUND

[0002] Temperature therapy has been used for many years. For example, ice baths are used to reduce the skin temperature of a user. The user may immerse a substantial portion of their body in cold water for at least a few minutes to reduce their skin temperature. The water of the ice bath may also apply pressure to the user's body to force blood towards the core of the user's body.

[0003] Setting up temperature therapy can be difficult. For example, an ice bath typically requires a vessel large enough to hold substantially all of the user's body, enough water to fill the vessel, and enough ice to cool the water and maintain a cold water temperature throughout the bath. Efforts have been made to improve the user experience. For example, inflatable ice vessels have been developed to reduce the storage space needed for the vessel and chiller devices have been developed to cool the water of the bath without the need to produce and add ice.

[0004] However, there remains a need for improved temperature therapy apparatus and methods.

SUMMARY

[0005] According to an aspect, there is provided a cold therapy shirt, comprising: a plurality of clothing panels forming a torso portion and first and second arms, the first and second arms each joined at upper ends thereof to the torso portion, and wherein each clothing panel extends from a first edge to a second edge opposite the first edge and each clothing panel includes an integrated flexible gel sheet to be cooled prior to a user wearing the cold therapy shirt, each integrated flexible gel sheet extending continuously across the clothing panel from the first edge to the second edge, and each integrated flexible gel sheet formed of a cold-therapy material having a semi-rigid form throughout a selected temperature use range.

[0006] In some examples, the cold therapy material is a hydrogel.

[0007] The selected temperature use range may be a range of 3 degrees Celsius to 20 degrees Celsius.

[0008] The plurality of clothing panels may include a first panel extending at least from a torso right side to a torso left side, and the continuous sheet of cold therapy material of the first panel also extends at least from the torso right side to the torso left side.

[0009] The first panel may extend at least from a lower end of the torso portion to an upper end of the torso portion, and the continuous sheet of cold therapy material of the first panel also extends at least from the lower end of the torso portion to the upper end of the torso portion.

[0010] The first panel may be a rear torso panel and the first edge is at a torso right side seam and the second edge is at a torso left side seam.

[0011] Each panel may comprise the sheet of cold therapy material enclosed between layers of fabric.

[0012] The cold therapy shirt may be sized and shaped to compress a torso and arms of the user. The plurality of clothing panels may be joined to one another using stretch stitching.

[0013] According to another aspect, there is provided cold therapy pants, comprising: one or more clothing panels forming first and second legs, the first and second legs joined at upper ends thereof via a waist portion, and wherein each clothing panel extends from a first edge to a second edge opposite the first edge and each clothing panel includes an integrated flexible gel sheet to be cooled prior to a user wearing the cold therapy pants, each integrated flexible gel sheet extending continuously across the clothing panel from the first edge to the second edge, and each integrated flexible gel sheet formed of a cold-therapy material having a semi-rigid form throughout a selected temperature use range.

[0014] In some examples, the one or more clothing panels includes a first panel, the first panel extending at least from a right side of one of the first and second legs to a left side of the one of the first and second legs, and the continuous sheet of cold therapy material of the first panel also extends at least from the right side of the one of the first and second legs to the left side of the one of the first and second legs.

[0015] The first panel may also extend at least from a lower end of the one of the first and second legs to the upper end of the one of the first and second legs, and the continuous sheet of cold therapy material of the first panel also extends at least from the lower end of the one of the first and second legs to the upper end of the one of the first and second legs.

[0016] The one or more clothing panels may include a plurality of clothing panels, and the first panel is a front panel and the first edge is at an inseam and the second edge is at an outseam.

[0017] Each panel may comprise the sheet of cold therapy material enclosed between layers of fabric.

[0018] The cold therapy pants may be sized and shaped to compress legs of the user. The one or more clothing panels may include a plurality of clothing panels joined to one another using stretch stitching.

[0019] According to another aspect, there is provided a method of cold therapy, comprising: cooling a cold therapy material in at least one cold therapy garment, the cold therapy material having a semi-rigid form throughout a selected temperature use range, the at least one cold therapy garment formed from one or more clothing panels, each clothing panel extending from a first edge to a second edge opposite the first edge and including a continuous flexible gel sheet formed of the cold-therapy material, each continuous flexible gel sheet extending across the panel from the first edge to the second edge; donning, after cooling the cold therapy material, the at least one cold therapy garment whereby the at least one cold therapy garment overlies at least 30% of a surface area of a wearer's body; wearing, after donning the at least one cold therapy garment, the at least one cold therapy garment for a period of time whereby a skin temperature of the wearer is decreased to a decreased temperature.

[0020] In some examples, the at least one cold therapy garment includes a cold therapy pants comprising first and second legs joined at upper ends thereof via a waist portion, the first and second legs formed from the one or more clothing panels.

[0021] The at least one cold therapy garment may be a plurality of cold therapy garments and further include a cold therapy shirt comprising a torso portion and first and second arms, the first and second arms each joined at upper ends thereof to the torso portion, the first and second arms and the torso portion formed from the clothing panels, and wherein donning the at least one cold therapy garment includes donning the cold therapy pants and the cold therapy shirt whereby the plurality of cold therapy garments overlies at least 60% of a surface area of the wearer's body.

[0022] Each of the at least one cold therapy garment may be a compression garment sized and shaped to compress an underlying body of the wearer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0023] The drawings included herewith are for illustrating various examples of systems, methods, and apparatus of the present specification. In the drawings:

[0024] FIG. 1 is a front view of a temperature therapy shirt, according to an embodiment;

[0025] FIG. 2 is a rear view of the temperature therapy shirt of FIG. 1;

[0026] FIG. 3 is a perspective cross sectional view of a clothing panel of the temperature therapy shirt of FIG. 1;

[0027] FIG. 4 is a diagram of stretch stitching examples;

[0028] FIG. 5 is a front view of a temperature therapy pants, according to an embodiment;

[0029] FIG. 6 is a rear view of the temperature therapy pants of FIG. 5;

[0030] FIG. 7 is a perspective cross sectional view of a clothing panel of another temperature therapy pants; and,

[0031] FIG. 8 is a flow diagram of a method of temperature therapy, according to an embodiment.

DETAILED DESCRIPTION

[0032] Various apparatus or processes will be described below to provide an example of each claimed embodiment. No example described below limits any claimed embodiment and any claimed embodiment may cover processes or apparatuses that differ from those described below. The claimed embodiments are not limited to apparatuses or processes having all of the features of any one apparatus or process described below or to features common to multiple or all of the apparatus or processes described below.

[0033] Referring now to FIGS. 1 and 2, illustrated therein is an exemplary temperature therapy garment **100**. A temperature therapy garment is heatable or coolable to be subsequently worn by a wearer to warm or cool, respectively, the wearer. The temperature therapy garment **100** is designed to simulate water therapy. For example, in cold water therapy, a user submerges at least a substantial portion of the user's body in cold water. The user may submerge at least 30% of their body (e.g., their lower body from about the waist down). The user may submerge at least 60% of their body (e.g., their lower body along with much of their torso and arms). A user may enjoy a temperature therapy treatment. The cold temperatures may help centralize blood flow into the core of the user's body. The warm temperatures may help with circulation in the user's body.

[0034] The temperature therapy garment **100** simulates water therapy by overlying a substantial part of the user's body (e.g., at least 30%) with a temperature therapy material. The temperature therapy material is integrated into the garment to overlie the user's body when the garment is worn. In use, the garment **100** is cooled (e.g., placed in a refrigerator or freezer or left outside during cold weather) or heated (e.g., placed in a microwave for a few seconds) and then donned by the wearer. In use, a cold therapy treatment using temperature therapy material overlying a substantial portion of the user's body (e.g., at least 30% or at least 60%) may result in brown fat activation, dopamine spikes (e.g., up to 250%), improved blood circulation and oxygen levels in the body (e.g., for easier waste and toxin removal from the body resulting in healthier organs, muscles and tissues).

[0035] Overlying a substantial portion of the user's body result in benefits that cannot be achieved through localized temperature treatment. For example, localized cold therapy may reduce inflammation and pain, but cannot achieve benefits from near full body cold exposure (e.g., as in ice bathing or cryotherapy).

[0036] The temperature therapy material integrated into the garment is formed into one or more flexible gel sheets. The sheet(s) of temperature therapy material have a semi-rigid form within an expected temperature use range. In some examples, the gel sheets of temperature therapy material are soft gel sheets within an expected temperature use range. In some examples, the temperature therapy material is a hydrogel. In some examples, the gel includes a polymer network. The expected temperature use range includes the range of temperatures from a temperature the garment is at while being prepared for use (e.g., a refrigerator temperature) to a room temperature. For a cold therapy material, the expected temperature use range may be between -20 and 30 degrees Celsius, -5 degrees Celsius and 30 degrees Celsius, -5 degrees Celsius and 25 degrees Celsius, -5 degrees Celsius and 20 degrees Celsius, 0 degrees Celsius and 20 degrees Celsius, or 3 degrees Celsius and 20 degrees Celsius. In some examples (e.g., when using certain freezers), when the material is prepared for use, a cold therapy material is expected to be cooled to between -40 and -1 degrees Celsius, -30 and -10 degrees Celsius, or -25 and -15 degrees Celsius, and so these temperatures are part of the expected temperature range. For a hot therapy material, the expected temperature use range may be between 15 degrees Celsius and 50 degrees Celsius, between 20 degrees Celsius and 45 degrees Celsius, or between 20 degrees Celsius and 40 degrees Celsius. It will be appreciated that the temperature therapy material may also be in the semirigid gel form outside the expected temperature use range.

[0037] The temperature therapy garment **100** includes one or more temperature therapy clothing panels **102**. Each panel **102** includes a sheet of the temperature therapy material. The sheet of temperature therapy material extends across substantially all of the panel **102**, e.g., to provide a

generally continuous coverage of the temperature therapy material over the body of the wearer. It will be appreciated that a temperature therapy clothing panel having a sheet of temperature therapy material extending across all of the panel may still include a short extent beyond the sheet of temperature therapy material, e.g., to facilitate joining the panel to an adjacent panel.

[0038] In some embodiments, a single panel **102** may be shaped into a garment. In other words, the garment may include only a single panel **102** shaped into a garment (e.g., forming one or more sleeves to receive arms or legs). However, in some embodiments the garment **100** includes a plurality of panels **102**, e.g., for ease of manufacturing. In embodiments which the garment **100** includes more than one panel **102**, panels **102** are joined to one another at edges **104** thereof. The panels **102** may be joined in any suitable way, such as by stitching or welding panels to one another. In some embodiments, the sheet of temperature therapy material extends up to the edge of the panel **102** to provide a generally continuous layer of the temperature therapy material overlying the wearer.

[0039] Referring now to FIG. 3, illustrated therein is a cross sectional view of an exemplary temperature therapy clothing panel **102**. The temperature therapy clothing panel **102** includes a sheet **106** of a temperature therapy material. The sheet **106** is a continuous sheet of temperature therapy material to provide a generally continuous coverage of temperature therapy material to the underlying body of the wearer when the garment is worn. The exemplary sheet **106** of temperature therapy material of FIG. 3 extends across substantially all of the panel **102** up to the edge **104** of the panel **102**. The exemplary panel **102** of FIG. 1 extends from a first edge **104a** to a second edge **104b**, and, as illustrated in FIG. 3, the exemplary sheet **106** extends across the panel **102** from the first edge to **104a** the second edge **104b**. In some examples, the sheet **106** is of a generally constant thickness across the panel **102**. The thickness **108** of the sheet **106** may be between 1 mm and 15 cm, between 1 mm and 10 cm, or between 1 mm and 5 cm.

[0040] The sheet **106** of temperature therapy material is integrated into the panel **102**. To cool or warm the temperature therapy material, the whole garment is cooled or warmed, rather than removing the temperature therapy material from the garment for cooling or warming. In some embodiments, the panel **102** includes a cover **110** enclosing the sheet **106**. The cover may protect the sheet **106** from damage and/or prevent direct user contact with the sheet **106** (e.g., from the sticky surfaces of the sheet if the sheet is formed of a sticky gel). The cover **110** permanently encloses the sheet **106**. The exemplary cover **110** of FIG. 3 includes fabric **112** enclosing the sheet **106**. The fabric **112** encases the sheet **106**. In some embodiments, the panel **102** includes an inner layer of fabric **112a** and an outer layer of fabric **112b** sandwiching the sheet **106**. The exemplary fabric layers **112a**, **112b** of FIG. 3 are stitched together to encase the sheet **106** of temperature therapy material. In other embodiments in which the cover **110** includes fabric, the fabric may be secured in other ways in addition to or in alternative to stitching (e.g., welding).

[0041] In some examples, the panel **102** is constructed to encourage easy thermal transfer from the sheet to an interior of the garment. In some examples, the cover **110** includes only a single layer over an inner side **106a** of the sheet **106**. In some examples, the panel **102** includes only the single layer of the cover between the inner side **106a** of the sheet **106** and an interior of the garment. A single layer reduces thermal insulation to allow easier heat transfer. In some examples, the cover **110** includes a single layer on each of the inner side **106a** and an opposite outer side **106b** of the sheet **106**. The exemplary cover **110** of FIG. 3 includes a single layer of fabric **112** on each side of the sheet **106**.

[0042] However, it will be appreciated that in some examples, the panel **102** and/or the cover **110** may include more than one layer on one or both sides of the sheet **106**. In some examples, an exterior side **106b** of the garment is insulated to reduce thermal transfer between the exterior side **106b** of the sheet and exterior of the garment. The panel **102** may include an insulating layer over the outer side **106b** of the sheet **106** in addition to a layer of the cover **102** (e.g., in addition to the outer fabric layer **112b**). In some examples, an additional garment may be worn over the garment

100 (e.g., a coat worn over the garment **100**) to reduce heat transfer from an exterior environment to the sheet **106**.

[0043] Referring again to FIGS. **1** and **2**, in some embodiments the garment **100** is a compression garment. A compression garment may help keep the temperature therapy material close to the wearer's skin. A compression garment may also help centralize blood flow to the core of the user's body. Compression simulates hydrostatic pressure acting upon a user during a temperature therapy bath. A compression garment fits snugly and has elastic properties. The garment **100** is sized and shaped to compress an underlying body part of a user. In some examples, the garment **100** also includes stretch stitching. The exemplary garment **100** of FIG. **1** includes stretch stitching **130** to join panels **102** to one another. The exemplary panel **102** of FIG. **3** includes stretch stitching **130** to join the upper and lower layers of fabric **112a**, **122b** together to enclose the sheet **106**. In some examples, the same stitching may be used to close a cover **110** of a panel **102** and to join panels **102** to one another. Any suitable stretch stitching may be used to allow the compression garment **100** to stretch. Referring to FIG. **4**, stretch stitching **130** may be or include, e.g., triple stretch stitching **130a**, stretch zigzag stitching **130b**, elastic stitching **130c**, stretch blind hem stitching **130d**, blind hem stitching **130e**, and/or elastic overlock stitching **130f**.

[0044] Referring again to FIGS. **1** and **2**, the exemplary garment **100** is a shirt. The shirt **100** includes a plurality of panels **102**. The panels **102** form a torso portion **120**, a first arm **122**, and a second arm **124**. The first arm **122** is joined at an upper end **126** thereof to the torso portion **120**. The second arm **124** is joined at an upper end **128** thereof to the torso portion **120**. It will be appreciated that the shirt may include one or more further portions **138**, such as a collar portion **138** joined to the torso portion **120**, a waist portion joined to the torso portion **120**, or a wrist portion joined to one of the arms **122**, **124**. The one or more further portions **138** are formed from panels **102** (e.g., formed from regular materials not incorporating temperature therapy material). In some examples, the torso portion **120**, the first arm **122**, and the second arm **124** consist of the panels **102** (e.g., as in the exemplary shirt of FIGS. **1** and **2**), optionally with one or more further portions **138**.

[0045] In some examples, the garment is formed from relatively large panels **102** to extend continuously across a substantial portion of a wearer's body. The exemplary garment **100** of FIG. **2** includes a first panel **102a** extending at least from a torso right side **142** to a torso left side **144**, and the continuous sheet **106** of temperature therapy material in the panel **102a** also extends at least from the torso right side **142** to the torso left side **144**. The exemplary first panel **102a** of FIG. **2** is a rear torso panel. The exemplary first edge **104a** of the first panel **102a** of FIG. **2** is at a torso right side seam **150** and the second edge **104b** of the first panel **102a** is at a torso left side seam **152**.

Optionally, the panel **102a** extends further than from the torso right side **142** to the torso left side **144**, such as also extending around from one of the front and back sides to the other. Optionally, the panel **102a** forms a sleeve encircling the torso and extending between an upper edge and a lower edge. However, in many embodiments the torso portion **120** includes at least a front panel and a back panel joined together to form the torso portion **120**, e.g., for ease of manufacturing.

[0046] In some examples, the first panel **102a** also extends from a lower end **146** of the torso portion **120** to an upper end **148** of the torso portion **120**, and the continuous sheet **106** of temperature therapy material in the panel **102a** also extends from the lower end **146** of the torso portion **120** to the upper end **148** of the torso portion **120**. The lower end of the first panel **102a** may form a bottom edge of the torso portion **120** and/or be joined to a further portion **138** of the garment **100**. The upper end of the first panel **102a** may form an upper edge of the torso portion **120** and/or be joined to a further portion **128** of the garment **100** (e.g., a collar portion).

[0047] The garment **100** is a reusable garment. In use, the garment (e.g., the temperature therapy material) is warmed or cooled, then worn by a wearer (e.g., for at least a few minutes), then removed. After the garment is removed, it can again be warmed or cooled, then worn by a wearer, and then removed, and the cycle can continue multiple times. For example, a user may place the garment in a refrigerator on a first day, take the garment out of the refrigerator and wear the

garment for five minutes on a second, subsequent day, place the garment in a closet on the second day after wearing the garment, move the garment into the refrigerator on a third, subsequent day, and then later on the third day take the garment out of the refrigerator and wear it for a few minutes.

[0048] While the exemplary garment **100** of FIGS. **1** and **2** is a shirt, the garment may take the form of another type of garment. In some examples, the garment **100** is a pair of pants. Referring now to FIGS. **5** and **6**, illustrated is an exemplary garment **200** formed as pants. Garment **200** is similar in some respects to garment **100** of FIGS. **1** and **2**, and similar features are indicated by similar reference characters incremented by **100**.

[0049] Garment **200** includes one or more clothing panels **202**. The clothing panel **202** is similar to the clothing panel **102**, including a sheet of temperature therapy material. The clothing panels **202** form a first leg **260** and a second leg **262**. The first leg **260** is joined at an upper end **264** thereof to a waist portion **266**. The second leg **262** is joined at an upper end **268** thereof to the waist portion **266**. In some examples, the first and second legs **260**, **262** consist of the one or more clothing panels **202**.

[0050] The garment **200** is a compression garment. The exemplary pants is a compression pants sized and shaped to compress the legs of a wearer. Stitching in the garment **200** may be compression stitching, such as to join panels **202** to one another.

[0051] The exemplary garment **200** includes a first panel **202b**. The first panel **202b** extends at least from a right side **270** of one of the first and second legs **260**, **262** to a left side **272** of the one of the first and second legs, and the continuous sheet of temperature therapy material of the first panel **202b** also extends at least from the right side **270** of the one of the first and second legs to the left side **272** of the one of the first and second legs. In some examples, the first panel **202b** extends beyond the first and/or second sides **270**, **272**, such as forming a circumferentially continuous sleeve to receive a leg therein. In some examples, the sheet of temperature therapy material also forms a circumferentially continuous sleeve. Referring to FIG. **7**, the exemplary first panel **202b'** extends around an interior with opposing edges **204'** of the panel **202b'** joined to one another to form a sleeve in which may be received, e.g., a leg. A continuous sheet **206'** of temperature therapy material of the exemplary panel **202b'** of FIG. **7** also extends around an interior of the sleeve.

[0052] However, in some examples, the pants include a plurality of clothing panels **202** joined to one another to form the legs. The exemplary first panel **202b** of FIGS. **5** and **6** is a front panel, and is joined to one or more additional panels **202** at an inseam **274** and an outseam **276** (e.g., joined to a cooperating back panel at the inseam and at the outseam).

[0053] In some examples, the first panel **202b** also extends at least from a lower end **280** of the one of the first and second legs to the upper end **282** of the one of the first and second legs, and the continuous sheet of temperature therapy material of the first panel also extends at least from the lower end **280** of the one of the first and second legs to the upper end **282** of the one of the first and second legs. The lower end of the first panel **202b** may form a bottom edge of the leg and/or be joined to a further portion of the garment (e.g., a hem portion). The upper end of the first panel **202b** may be joined directly to the waist portion **266** or continue through the waist portion **266** and form the upper edge of the pants and/or be joined to a further portion **238** of the garment (e.g., a waistband). In some examples, the waist portion **266** is also formed of the one or more clothing panels **202**.

[0054] In some embodiments, more than one garment **100** is provided for use together. In some embodiments, a temperature therapy suit is provided. The suit covers more than 50% of the user's body. In some examples, the suit covers more than 60% of the user's body. The suit includes a shirt and a pair of pants, such as the shirt of FIGS. **1** and **2** and the pants of FIGS. **5** and **6**.

[0055] Referring now to FIG. **8**, illustrated is an exemplary method of temperature therapy **300**. Method **300** includes, at step **302**, cooling or warming a temperature therapy material in at least one temperature therapy garment. The at least one temperature therapy garment is formed from one or

more clothing panels. Each clothing panel extends from a first edge to a second edge opposite the first edge and includes a continuous sheet of the temperature therapy material extending across the panel from the first edge to the second edge, the temperature therapy material having a gel form within an expected use range. In some examples, the expected use range for cold therapy is any of the ranges described herein, such as at least a temperature range of -20 degrees Celsius to 30 degrees Celsius or 3 degrees Celsius to 20 degrees Celsius. In some examples, the expected use range for hot therapy is any of the ranges described herein, such as at least a temperature range of 20 degrees Celsius to 35 degrees Celsius. In some examples, the garment is the exemplary garment **100** of FIGS. **1** and **2** or the exemplary garment **200** of FIGS. **5** and **6**. In some examples, step **302** includes cooling or warming the temperature therapy material in more than one garment. In some examples, the at least one garment is a compression garment sized and shaped to compress an underlying body of a wearer.

[0056] At step **304**, method **300** includes donning, after cooling or warming the temperature therapy material, the at least one temperature therapy garment whereby the at least one temperature therapy garment overlies at least 30% of a surface area of a wearer's body. In some examples, step **304** includes donning more than one temperature therapy garment. In some examples, step **304** includes donning one or more temperature therapy garments such that the garments overly at least 60% of a surface area of a wearer's body. Method **300** also includes, at step **306**, wearing, after donning the at least one temperature therapy garment, the at least one temperature therapy garment for a period of time whereby a skin temperature of the wearer is changed to a target temperature. In some examples, for cold therapy the target temperature is a decreased temperature. The decreased temperature may be less than 30 degrees Celsius, less than 27 degrees Celsius, or less than 25 degrees Celsius. In some examples, for hot therapy the target temperature is an increased temperature. The increased temperature may be more than 37 degrees Celsius, more than 40 degrees Celsius, or more than 43 degrees Celsius.

[0057] At step **308**, method **300** includes removing the at least one temperature therapy garment after the skin temperature of the wearer decreases to the decreased temperature.

EXAMPLES

[0058] The effects of using cold therapy clothing panels (such as exemplary panel **102** of FIGS. **1** to **3**) were compared to the effects of an ice bath. In each cold therapy clothing panel treatment, the cold therapy clothing panels were cooled prior to use in a refrigerator to about -18 degrees Celsius. Example 1

[0059] A subject took a five minute ice bath in the morning using 27 pounds of ice. The subject's skin temperature immediately following the ice bath was measured at between 24 and 25 degrees Celsius. In the afternoon of the same day, the subject placed cold therapy clothing panels on the subject's thighs. The subject's skin temperature immediately following the cold therapy clothing panel treatment was measured at 25 degrees Celsius.

Example 2

[0060] On a first day, a subject took a seven minute ice bath using 20 pounds of ice. The subject's skin temperature immediately following the ice bath was measured at between 24 and 26 degrees Celsius.

[0061] On a second day, the subject took a five minute cold shower in the morning. The subject's skin temperature immediately following the cold shower was measured at between 28 and 29 degrees Celsius. In the afternoon of the second day, the subject placed cold therapy clothing panels on the subject's thighs for three minutes. The subject's skin temperature immediately following the cold therapy clothing panel treatment measured at between 22 and 23 degrees Celsius.

Example 3

[0062] On a first day, a subject took a four minute ice bath using 53 pounds of ice with a water temperature of 9.5 degrees Celsius. The subject's skin temperature immediately following the ice bath was measured at between 22 and 23 degrees Celsius.

[0063] On a second day, the subject took a six minute ice bath using 27 pounds of ice with a water temperature of 15 degrees Celsius. The subject's skin temperature immediately following the ice bath was measured at between 24 and 25 degrees Celsius.

[0064] On a third day, the subject took a six minute ice bath following a regular temperature shower, the ice bath using 27 pounds of ice and with a water temperature of 15 degrees Celsius. The subject's skin temperature immediately following the ice bath was measured at 24 degrees Celsius.

[0065] On a fourth day, the subject took a six minute ice bath following a regular temperature shower, the ice bath using 27 pounds of ice and with a water temperature of 14 degrees Celsius. The subject's skin temperature immediately following the ice bath was measured at between 22 and 23 degrees Celsius.

[0066] On a fifth day, the subject took a regular temperature shower and, immediately following the shower, the subject placed cold therapy clothing panels on the subject's thighs and cold therapy clothing panels on the subject's arms for four minutes. The subject's skin temperature immediately following the cold therapy clothing panel treatment was measured at between 22 and 24 degrees Celsius.

[0067] The present invention has been described here by way of example only. Various modification and variations may be made to these examples without departing from the scope of the invention, which is limited only by the appended claims.

Claims

1. A cold therapy shirt, comprising: a plurality of clothing panels forming a torso portion and first and second arms, the first and second arms each joined at upper ends thereof to the torso portion, and wherein each clothing panel extends from a first edge to a second edge opposite the first edge and each clothing panel includes an integrated flexible gel sheet to be cooled prior to a user wearing the cold therapy shirt, each integrated flexible gel sheet extending continuously across the clothing panel from the first edge to the second edge, and each integrated flexible gel sheet formed of a cold-therapy material having a semi-rigid form throughout a selected temperature use range.
2. The cold therapy shirt of claim 1, wherein the cold therapy material is a hydrogel.
3. The cold therapy shirt of claim 1, wherein the selected temperature use range includes a range of 3 degrees Celsius to 20 degrees Celsius.
4. The cold therapy shirt of claim 1, wherein the plurality of clothing panels includes a first panel extending at least from a torso right side to a torso left side, and the continuous sheet of cold therapy material of the first panel also extends at least from the torso right side to the torso left side.
5. The cold therapy shirt of claim 4, wherein the first panel also extends at least from a lower end of the torso portion to an upper end of the torso portion, and the continuous sheet of cold therapy material of the first panel also extends at least from the lower end of the torso portion to the upper end of the torso portion.
6. The cold therapy shirt of claim 5, wherein the first panel is a rear torso panel and the first edge is at a torso right side seam and the second edge is at a torso left side seam.
7. The cold therapy shirt of claim 1, wherein each panel comprises the sheet of cold therapy material enclosed between layers of fabric.
8. The cold therapy shirt of claim 1, wherein the cold therapy shirt is sized and shaped to compress a torso and arms of the user and the plurality of clothing panels are joined to one another using stretch stitching.
9. Cold therapy pants, comprising: one or more clothing panels forming first and second legs, the first and second legs joined at upper ends thereof via a waist portion, and wherein each clothing panel extends from a first edge to a second edge opposite the first edge and each clothing panel includes an integrated flexible gel sheet to be cooled prior to a user wearing the cold therapy pants,

each integrated flexible gel sheet extending continuously across the clothing panel from the first edge to the second edge, and each integrated flexible gel sheet formed of a cold-therapy material having a semi-rigid form throughout a selected temperature use range.

10. The cold therapy pants of claim 9, wherein the cold therapy material is a hydrogel.

11. The cold therapy pants of claim 9, wherein the selected temperature use range includes a range of 3 degrees Celsius to 20 degrees Celsius.

12. The cold therapy pants of claim 9, wherein the one or more clothing panels includes a first panel, the first panel extending at least from a right side of one of the first and second legs to a left side of the one of the first and second legs, and the continuous sheet of cold therapy material of the first panel also extends at least from the right side of the one of the first and second legs to the left side of the one of the first and second legs.

13. The cold therapy pants of claim 12, wherein the first panel also extends at least from a lower end of the one of the first and second legs to the upper end of the one of the first and second legs, and the continuous sheet of cold therapy material of the first panel also extends at least from the lower end of the one of the first and second legs to the upper end of the one of the first and second legs.

14. The cold therapy pants of claim 13, wherein one or more clothing panels includes a plurality of clothing panels, and the first panel is a front panel and the first edge is at an inseam and the second edge is at an outseam.

15. The cold therapy pants of claim 9, wherein each panel comprises the sheet of cold therapy material enclosed between layers of fabric.

16. The cold therapy pants of claim 9, wherein the cold therapy pants is sized and shaped to compress legs of the user, and the one or more clothing panels includes a plurality of clothing panels joined to one another using stretch stitching.

17. A method of cold therapy, comprising: cooling a cold therapy material in at least one cold therapy garment, the cold therapy material having a semi-rigid form throughout a selected temperature use range, the at least one cold therapy garment formed from one or more clothing panels, each clothing panel extending from a first edge to a second edge opposite the first edge and including a continuous flexible gel sheet formed of the cold-therapy material, each continuous flexible gel sheet extending across the panel from the first edge to the second edge; donning, after cooling the cold therapy material, the at least one cold therapy garment whereby the at least one cold therapy garment overlies at least 30% of a surface area of a wearer's body; wearing, after donning the at least one cold therapy garment, the at least one cold therapy garment for a period of time whereby a skin temperature of the wearer is decreased to a decreased temperature.

18. The method of claim 17, wherein the at least one cold therapy garment includes a cold therapy pants comprising first and second legs joined at upper ends thereof via a waist portion, the first and second legs formed from the one or more clothing panels.

19. The method of claim 18, wherein the at least one cold therapy garment is a plurality of cold therapy garments and further includes a cold therapy shirt comprising a torso portion and first and second arms, the first and second arms each joined at upper ends thereof to the torso portion, the first and second arms and the torso portion formed from the clothing panels, and wherein donning the at least one cold therapy garment includes donning the cold therapy pants and the cold therapy shirt whereby the plurality of cold therapy garments overlies at least 60% of a surface area of the wearer's body.

20. The method of claim 17, wherein each of the at least one cold therapy garment is a compression garment sized and shaped to compress an underlying body of the wearer.
